

Hypertension & Heart Disease Prediction Using Machine Learning

Srikanta Saha

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1 Introduction

This project aims to develop machine learning models to predict hypertension and heart disease using a healthcare dataset. The dataset includes age, gender, marital status, employment status, place of residence, BMI, glucose level, and smoking status. The stroke variable is excluded from the analysis. The dataset contains a total of 5110 observations

The variables are described below:

- Age – Continuous
- Gender – Categorical (Male, Female, Others)
- Ever Married – Categorical (No, Yes)
- Employment Status – Categorical (Children, Never Worked, Self-employed, Govt. Job, Private)
- Place of Residence – Categorical (Rural, Urban)
- Smoking Status – Categorical (Unknown, Never Smoked, Formerly Smoked, Currently Smoking)
- BMI – Continuous
- Average Glucose Level – Continuous

The target variables for classification are:

- Hypertension (0 = No, 1 = Yes)
- Heart Disease (0 = No, 1 = Yes)

2 Data Preprocessing

Several preprocessing steps are performed. Initially, the id and stroke variables are removed from the dataset.

The Body Mass Index (BMI) variable contains missing values represented as “N/A”. These observations are treated as missing values and excluded from the analysis. After removing missing BMI observations, the final analytical dataset contains 4,908 observations.

The gender category “Other” contains only one observation and is removed as an outlier observation.

BMI is categorized into four groups based on WHO criteria: underweight (< 18.5), normal ($18.5 - 22.9$), overweight ($23 - 24.9$), and obese (≥ 25). Cross-tabulations are subsequently examined to understand the distribution of hypertension across other variables and to study the relationships between them.

3 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is performed to understand the structure, distribution and relationships among variables in the dataset before model development.

The prevalence of hypertension is slightly higher among males (9.95%) compared to females (8.66%). A similar pattern is observed for heart disease, where males show a substantially higher prevalence (7.11%) than females (3.45%). These results suggest that gender may play an important role in cardiovascular risk, with males appearing more vulnerable to both hypertension and heart disease in this dataset.

Table-1 shows that the prevalence of both hypertension and heart disease increases substantially with age. Hypertension prevalence rises from 0% in the 0–16 age group to 23.43% in the 76–90 age group. Similarly, heart disease prevalence increases from 0.13% in younger age groups to 19.72% among individuals aged 76–90 years. These results suggest that ageing is strongly associated with increased cardiovascular risk and highlights age as one of the most important predictors in the classification models.

Table 1: Age-wise Distribution of Hypertension and Heart Disease

Age Group	Total Sample	Hypertension		Heart Disease	
		Cases	%	Cases	%
0–16	778	0	0.00	1	0.13
17–30	757	10	1.32	1	0.13
31–45	1011	46	4.55	4	0.40
46–60	1145	140	12.23	54	4.72
61–75	786	154	19.59	98	12.47
76–90	431	101	23.43	85	19.72

The boxplot of BMI VS Hypertension indicates that most observations are concentrated within a moderate BMI range, with a median BMI near 28. The distribution appears positively skewed due to several high-value outliers.

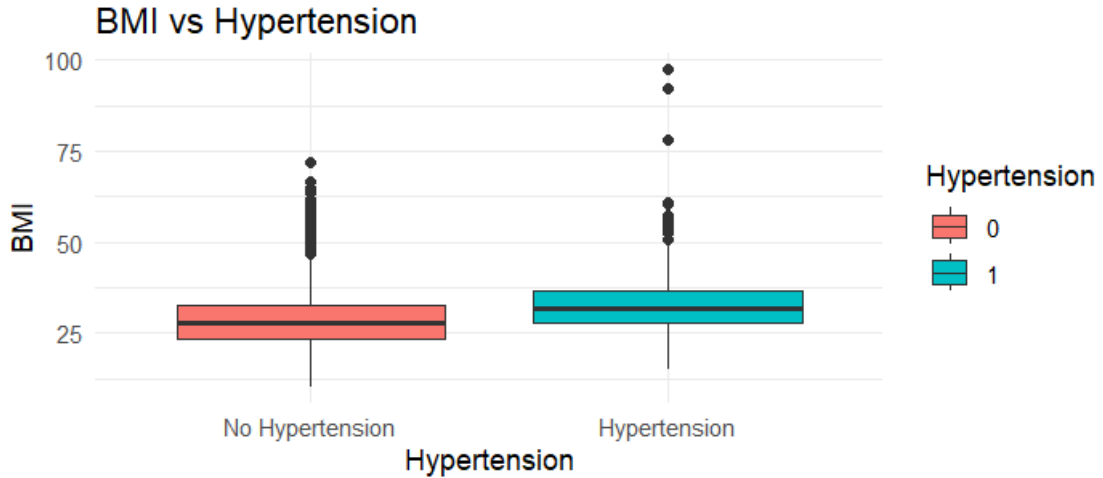


Figure 1: Box Plot BMI VS Hypertension

The boxplot (Fig-2) compares BMI between patients with and without heart disease. Both groups have a similar median BMI of around 28-29, meaning BMI is almost the same for both groups. The "No Heart Disease" group has more extreme outliers, with some patients having very high BMI values up to 90, while the "Heart Disease" group is more compact. This suggests that BMI alone is not a strong predictor of heart disease in this dataset, and other factors like age and glucose level likely play a bigger role.



Figure 2: Box Plot BMI VS Heart Disease

The boxplots (Fig-3 , Fig-4) indicate that individuals with hypertension and heart disease generally had higher average glucose levels compared to individuals without these conditions. In both plots, the median glucose level was noticeably higher among diseased individuals, and the spread of glucose values was also wider, suggesting greater variability in glucose concentration among high-risk groups. Several extreme glucose values were observed in both disease categories, indicating the presence of individuals with substantially elevated glucose levels. These results suggest that elevated average glucose level is strongly associated with both hypertension and heart disease and may serve as an important predictor in cardiovascular risk classification models.

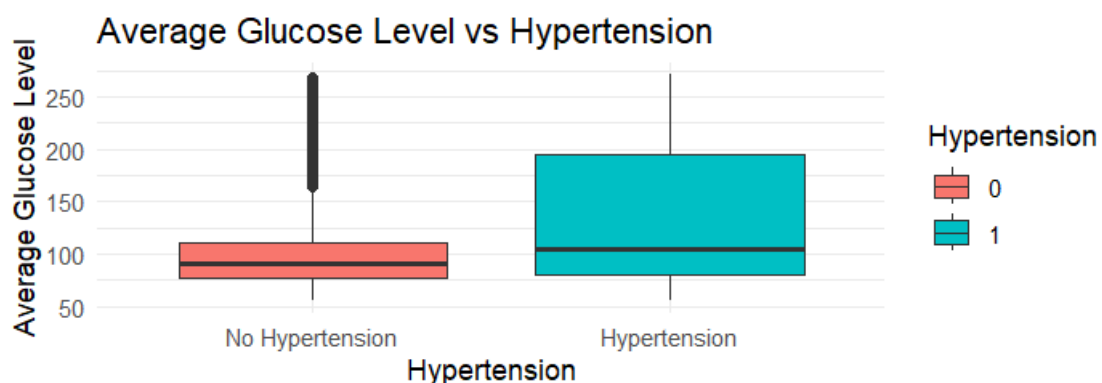


Figure 3: Box Plot Hypertension VS Average Glucose Level

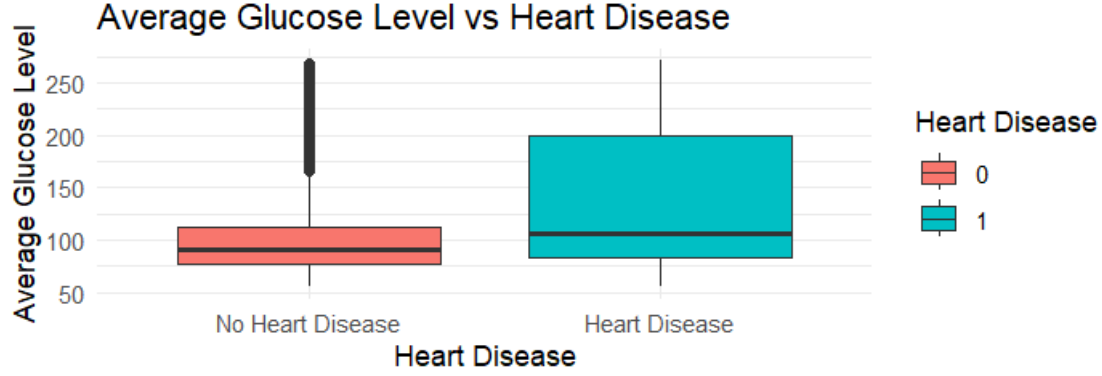


Figure 4: Box Plot Heart Disease VS Average Glucose Level

4 Data Splitting

After data preprocessing, the dataset is divided into training and testing datasets using an 80:20 ratio. The training dataset is used for model development, while the testing dataset is used to evaluate predictive performance on unseen observations.

A stratified sampling approach is applied to ensure that the distribution of hypertension classes remained approximately similar in both datasets.

5 Class Imbalance Handling

Class imbalance can bias machine learning models toward the majority class, leading to poor detection of minority cases. To address this issue, a data balancing technique is applied using the `ROSE` package in R. Oversampling is performed to increase the number of minority class observations in the training dataset, thereby improving the model's ability to correctly identify hypertension and heart disease cases.

6 Machine Learning Models

This study uses eight classification ML algorithms: Logistic Regression, Random Forest, SVM, Gradient Boosting, Naïve Bayes, AdaBoost, XGBoost and LDA.

A set of performance metrics are documented for each model that specifies Test data Accuracy, area under the ROC curve (AUC), F1 score, precision, recall. The evaluation process utilises an 80-20 divide test strategy where 80% of the dataset is reserved for training the algorithm, while 20% is used for evaluating the model's performance.

Important variables for each classifier are identified using model-specific variable importance measures. For Logistic Regression, the magnitude and significance of model coefficients are used to determine influential predictors. In Random Forest, variables are ranked based on Mean Decrease Accuracy and Mean Decrease Gini values. For Gradient Boosting and XGBoost, feature importance scores are obtained from the relative influence and gain measures generated by the models. These approaches help identify

the variables that contributed most to predicting hypertension and heart disease across different machine learning algorithms.

6.1 Hypertension Classifier Performance

For hypertension classification, XGBoost achieves the highest AUC (0.831), indicating the strongest ability to discriminate between hypertensive and non-hypertensive individuals across thresholds. Logistic Regression and AdaBoost show slightly better F1 scores and balanced overall performance, while SVM achieves a very high recall (0.856). Random Forest reports the highest accuracy (0.863), but its recall is extremely low (0.278), meaning that it fails to identify many true hypertension cases. This demonstrates that high accuracy can be misleading in imbalanced datasets because the majority class dominates predictions.

Table 2: Performance Comparison of Hypertension Classification Models

Model	Accuracy	Precision	Recall	F1 Score	AUC
XGBoost	0.667	0.198	0.867	0.323	0.831
Gradient Boosting	0.690	0.201	0.800	0.321	0.818
LDA	0.699	0.203	0.778	0.322	0.815
Logistic Regression	0.705	0.208	0.789	0.329	0.813
AdaBoost	0.721	0.214	0.767	0.335	0.809
SVM	0.695	0.212	0.856	0.340	0.805
Naive Bayes	0.652	0.185	0.822	0.302	0.797
Random Forest	0.863	0.266	0.278	0.272	0.793

Table 3: Important Variables Identified by Different Hypertension Classification Models

Model	Important Variables
Logistic Regression	Gender, Age, Average Glucose Level, BMI, Smoking Status
Random Forest	Age, Average Glucose Level, Smoking Status
Gradient Boosting	Age, Average Glucose Level, Smoking Status
XGBoost	Age, Average Glucose Level, BMI, Smoking Status

6.2 Heart Disease Classifier Performance

A similar pattern is observed in the heart disease classification exercise. LDA and Logistic Regression achieve the best AUC values (0.898 and 0.896 respectively), indicating excellent discriminative ability. Random Forest again shows the highest accuracy (0.934), but its recall is only 0.250, meaning that most true heart disease cases are missed. In contrast, Logistic Regression, LDA, and Naive Bayes achieved very high recall values above 0.87, making them more suitable for screening purposes where identifying high-risk individuals is more important than maximizing overall accuracy.

Table 4: Performance Comparison of Heart Disease Classification Models

Model	Accuracy	Precision	Recall	F1 Score	AUC
LDA	0.764	0.162	0.917	0.276	0.898
Logistic Regression	0.781	0.170	0.896	0.286	0.896
XGBoost	0.753	0.149	0.854	0.254	0.887
Gradient Boosting	0.763	0.152	0.833	0.257	0.885
SVM	0.778	0.160	0.833	0.268	0.877
AdaBoost	0.840	0.189	0.688	0.296	0.875
Naive Bayes	0.697	0.126	0.875	0.221	0.856
Random Forest	0.934	0.293	0.250	0.270	0.851

Table 5: Important Variables Identified by Different Heart Disease Classification Models

Model	Important Variables
Logistic Regression	Gender, Age, Employment Status, Average Glucose Level, Smoking Status
Random Forest	Age, Average Glucose Level, Smoking Status
Gradient Boosting	Age, Average Glucose Level, Gender
XGBoost	Age, Average Glucose Level, Gender

6.3 ROC Curve Comparison

The ROC curves further support these results. Models with higher AUC values, especially XGBoost for hypertension and Logistic Regression/LDA for heart disease, showed better separation between positive and negative classes across thresholds. Their ROC curves would lie closer to the upper-left corner, indicating stronger sensitivity-specificity balance. Random Forest produces comparatively lower AUC values despite high accuracy, proving that it is conservative in predicting positive cases and less effective in distinguishing diseased individuals consistently.

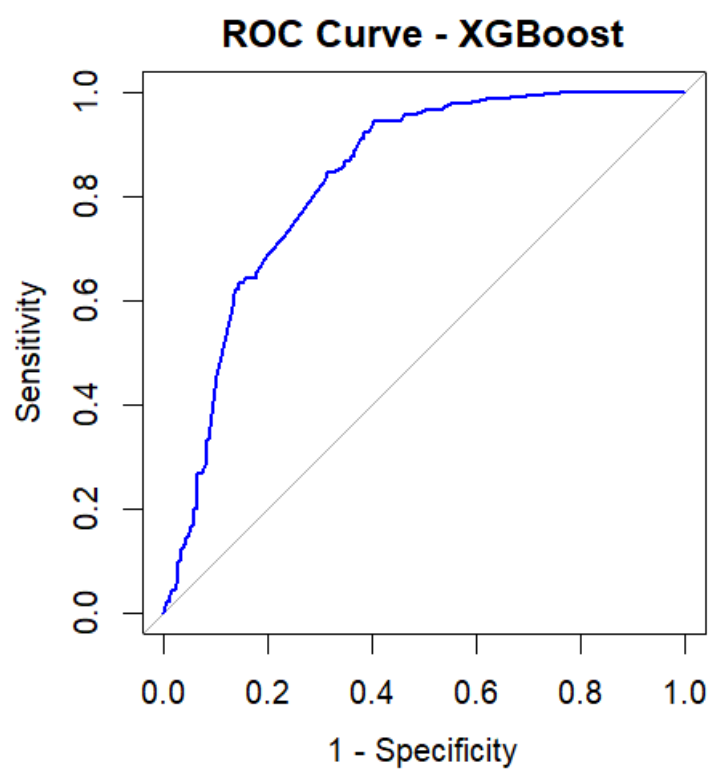


Figure 5: Hypertension- ROC Curve (XGBoost)

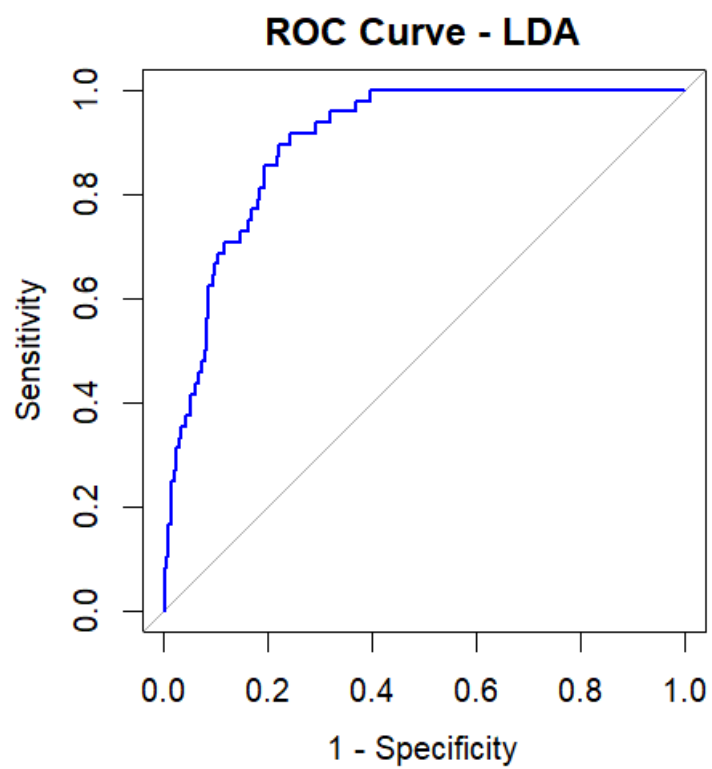


Figure 6: Heart Disease- ROC Curve (LDA)

6.4 Practical Model Deployment

In the practical setting, XGBoost should be preferred if the primary goal is maximising predictive performance. It achieves the highest AUC in both hypertension and heart disease classification and correctly identifies most positive cases through high recall. XGBoost can also capture nonlinear relationships and interactions between variables that other models miss.

6.5 Joint Predicted Risk Analysis: Hypertension and Heart Disease

To find whether individuals predicted to have hypertension are also predicted to have heart disease, both models are tested on the same set of patients. The data are first split into 80% for training and 20% for testing, and two separate training sets are prepared—one for predicting hypertension and one for predicting heart disease—using logistic regression with the same eight input variables: gender, age, marital status, work type, residence type, glucose level, BMI category, and smoking status. For each patient in the test set, two probabilities are obtained—the probability of having hypertension and the probability of having heart disease—and these are plotted against each other in a scatter plot. The points follow a clear diagonal pattern from the bottom-left to the upper-right corner, indicating that patients with a high predicted probability of hypertension also tend to have a high predicted probability of heart disease, and vice versa. This pattern is expected because both conditions share common underlying risk factors such as older age, elevated glucose level, and high BMI, which increase the predicted risk for both diseases simultaneously. The “Both” group (red points) and “HD only” group (blue points) appear mainly in the upper-right region of the plot, indicating high predicted risk on both axes, while the “Neither” group (grey points) is concentrated in the lower-left region, indicating low predicted risk for both conditions.

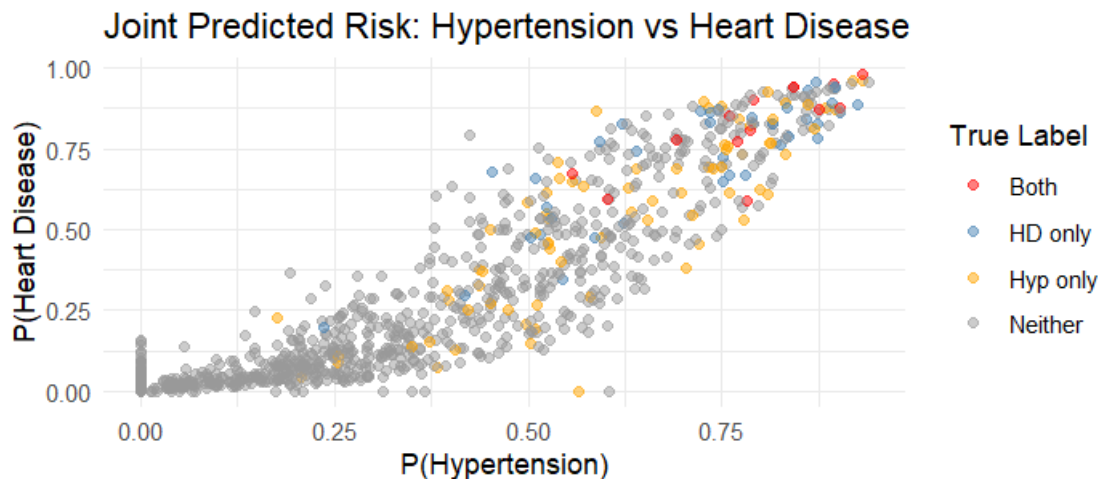


Figure 7: Joint Predicted Risk: Hypertension vs Heart Disease; HD: Heart Disease , Hyp: Hypertension